

香港中文大學(深圳)
The Chinese University of Hong Kong, Shenzhen



人工智能學院
School of Artificial Intelligence

AIE1901-S1 Lifelong AI: Lecture 1: Introduction

Yang Li

2026/9/7

Today's Plan

- Course Information
- Introduction to Lifelong AI

About this Course

- Course Website: <https://yangli-feasibility.com/home/classes/aie2026fall/>
- Introduction to Lifelong AI

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Course code	AIE1901
Course title	AI Exploration I
Session	01: AI That Learns Throughout Its Lifetime
Units	2
Instructional hours	2 hours per week (lecture and lab)
Meeting time	Monday, 10:30am-12:20pm
Locations	Lecture: TA301; Lab: C501

Today's Plan

- **Course Format:** Alternate between lecture and in-class labs (meeting location will depend on the schedule.)
- **Lab & Homeworks**
 - Work in groups of 2 people
 - Guided lab + open questions, submitted in written reports
 - **Course project:** select a Lifelong learning application and progressively explore how to make it learn better
 - **Midterm & Final:** group presentations to show progress

AI Exploration Dean's Award

- Each year the School of AI will select best projects from course AIE 1901-1904
- How to win this award
 - Choose a meaningful project (be creative!)
 - Good presentation and documentation
 - Solve technical challenge
 - Balanced team contribution

Grading

Component / Method	Weight
Assignments	40%
Middle-term Assessment	30%
Final-term Assessment	30%
Total	100%

Honor Code & AI Policy

- Must complete all coursework and assessments independently, unless AI tool use is explicitly permitted
- When allowed, give explicit **acknowledgement** on resources and explain how they were used

Next: what we will learn in this course and why?

Introduction to Lifelong AI

- How learning works in artificial intelligence
- How does it differ from human learning?
- What is lifelong learning?

What is Intelligence?

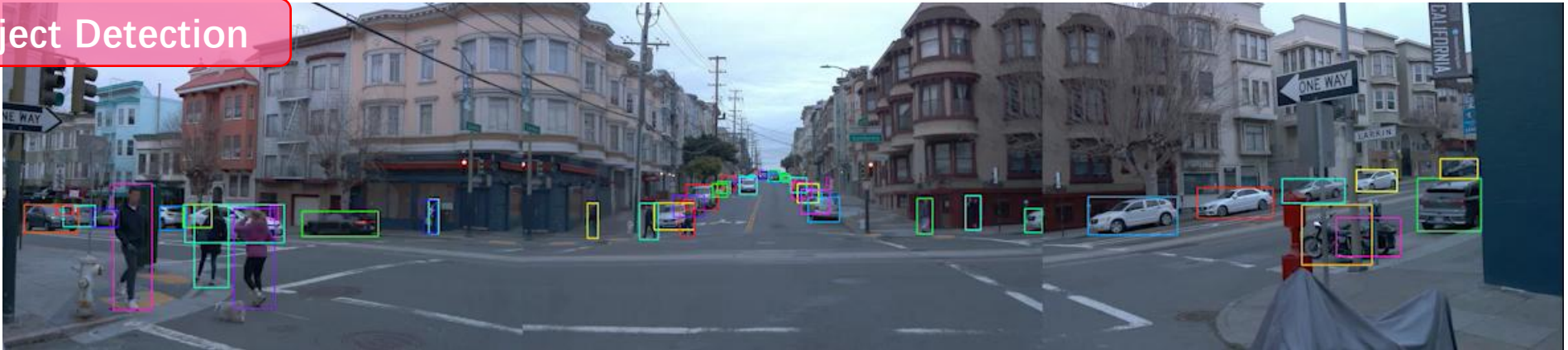
- Sensory Perception
 - Vision
 - Audio
 - Touch
- Language
- Memory
- Reasoning
- ...



2D/3D Visual Perception

- Example: autonomous driving perception

Object Detection

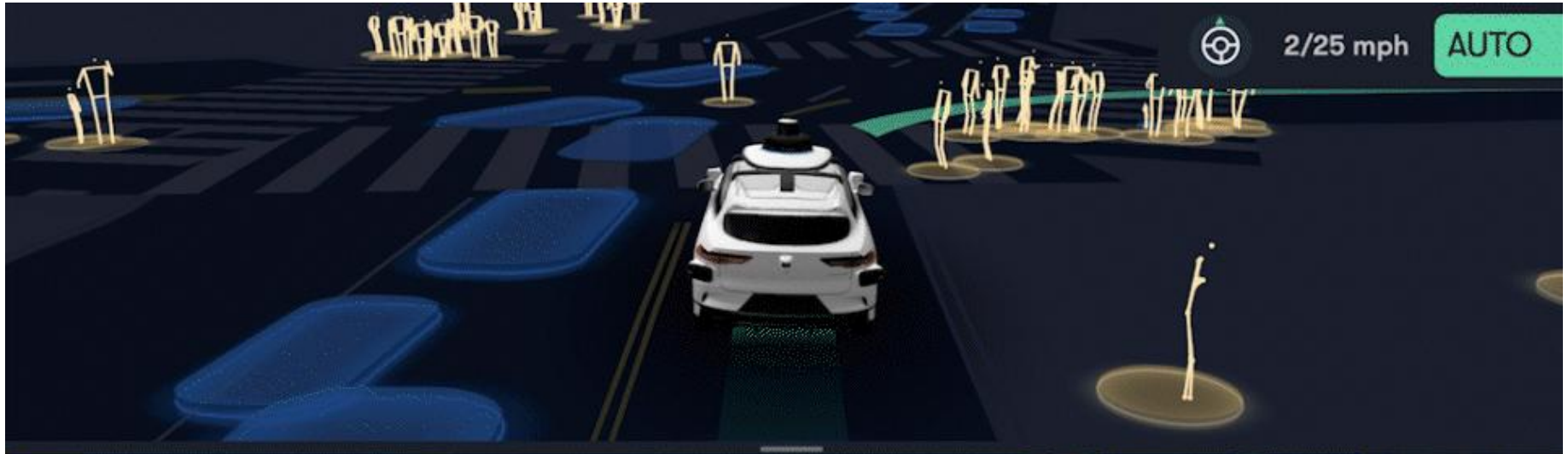


Semantic Segmentation



Waymo Open Dataset-Perception

2D/3D Visual Perception



Pose Estimation

Waymo Open Dataset-Perception

Language & Reasoning

Example: Generative AI

Sentence completion

- When the cat knocked over my water glass, the keyboard got ____
- A shipping container can hold 436 twelve-packs or 240 24-packs, so it's better to use _____



- When the cat knocked over my water glass, the keyboard got **wet**.
- A shipping container can hold 436 twelve-packs or 240 24-packs, so it's better to use **24-packs**.



Language & Reasoning

Global (International)



ChatGPT

OpenAI



Claude

Anthropic

China (Mainland)



豆包
Doubao

ByteDance
(字节跳动)



通义千问
Qwen

Alibaba
(阿里巴巴)



文心一言
Wenxin Yiyan

Baidu
(百度)



月之暗面
Moonshot AI
(月之暗面)



智谱AI
Zhipu AI
(智谱AI)

LaMDA, from Google Research, 2021



腾讯
Tencent
(腾讯)



MINIMAX

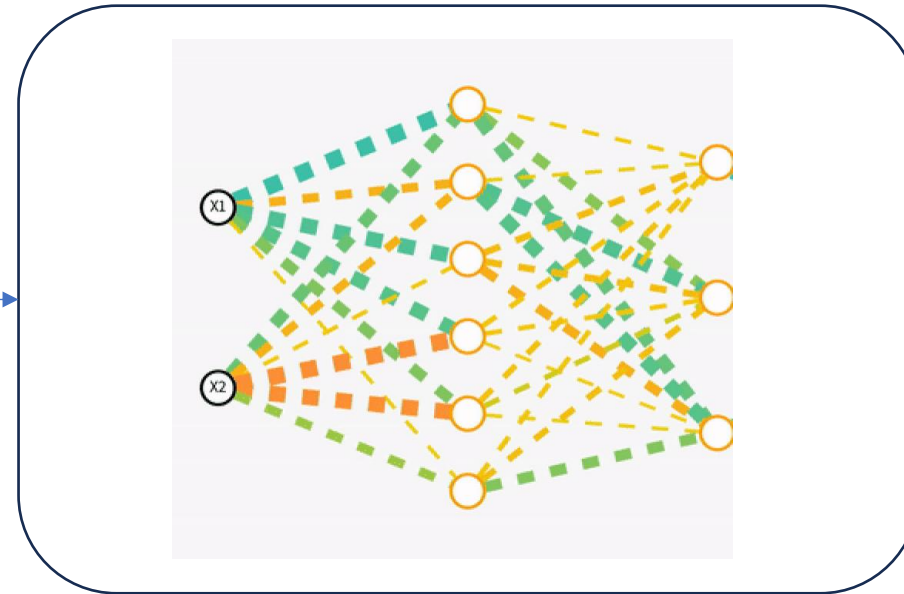
MiniMax
(稀宇科技)

AI is a Learning Machine

Example: Hand gesture classification



Input

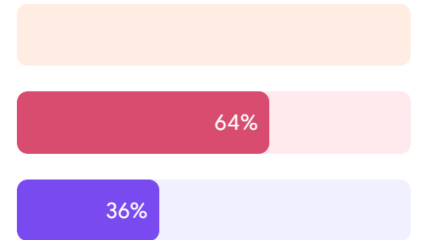


Neural Network

Rock

Paper

Sciss...



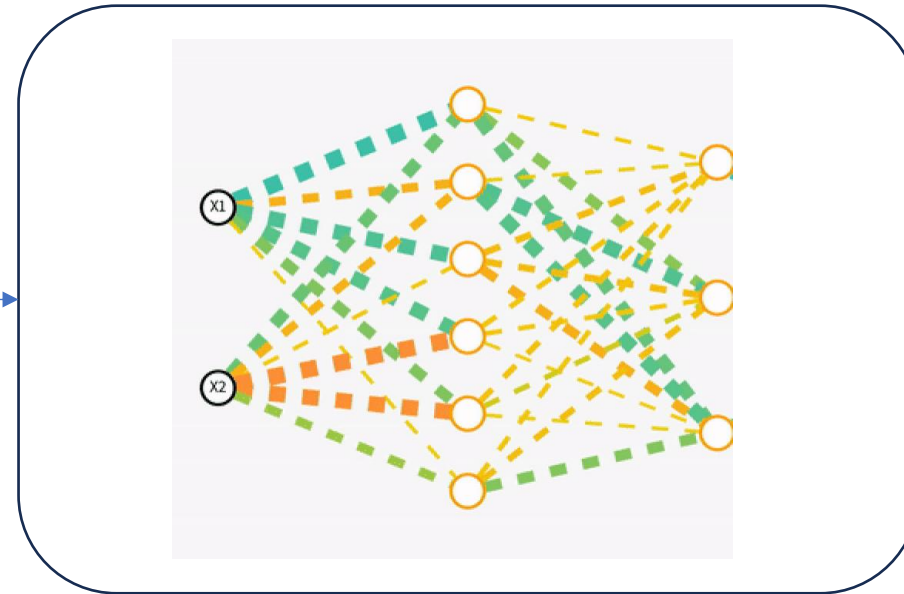
Output

AI is a Learning Machine

Example: Hand gesture classification

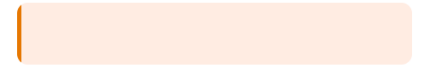


Input



Neural Network

Rock



Paper



Sciss...



Output

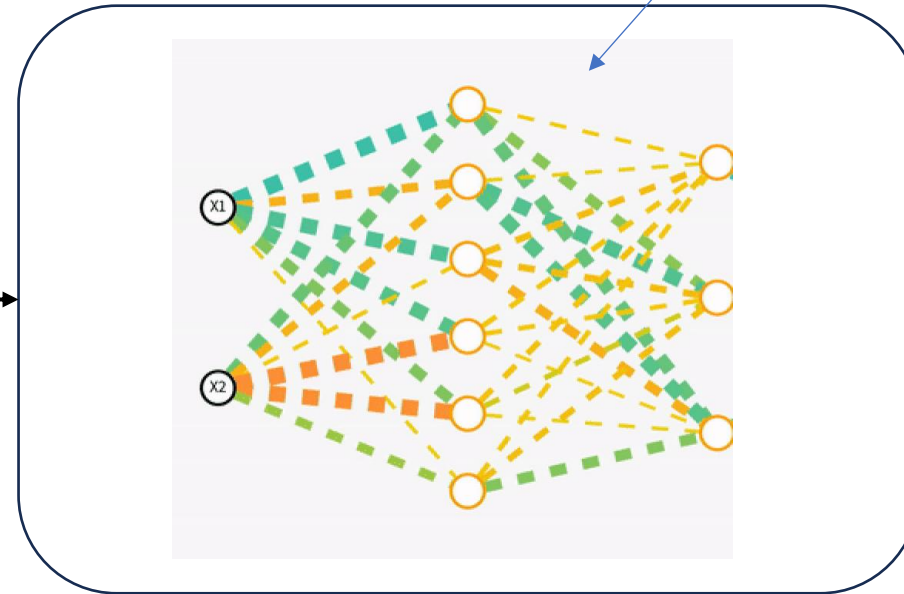
AI is a Learning Machine

Example: Hand gesture classification

Connection weights controls how an input maps to the output



Input



Neural Network

Rock

79%

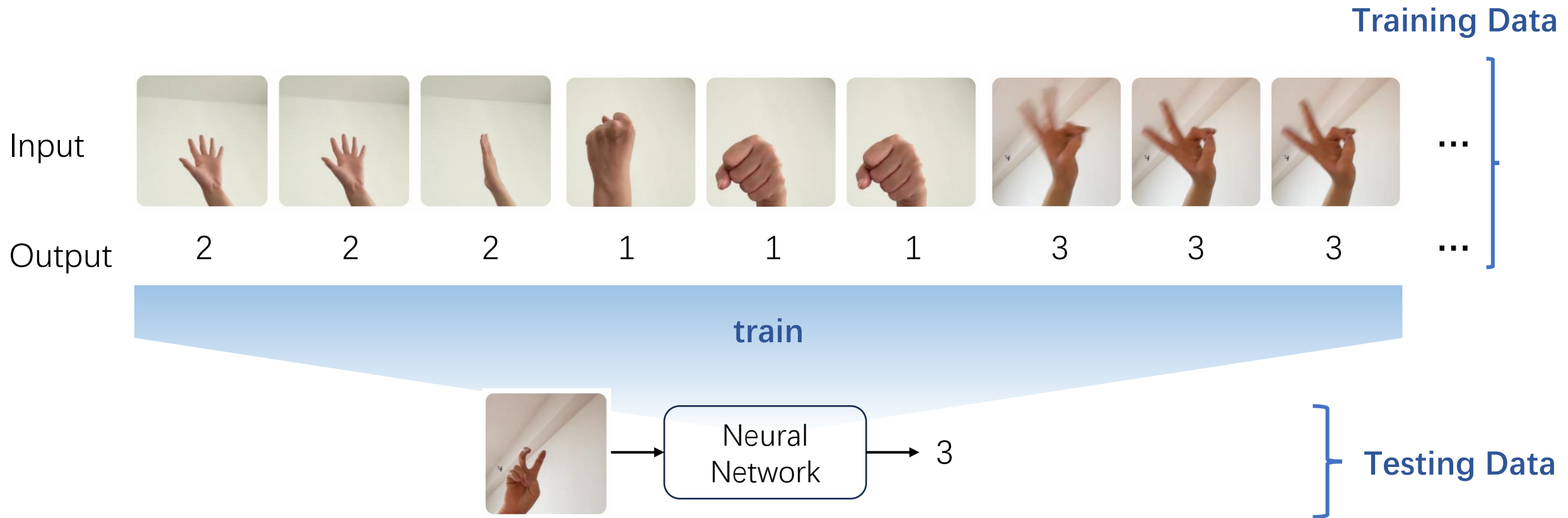
Paper

Sciss...

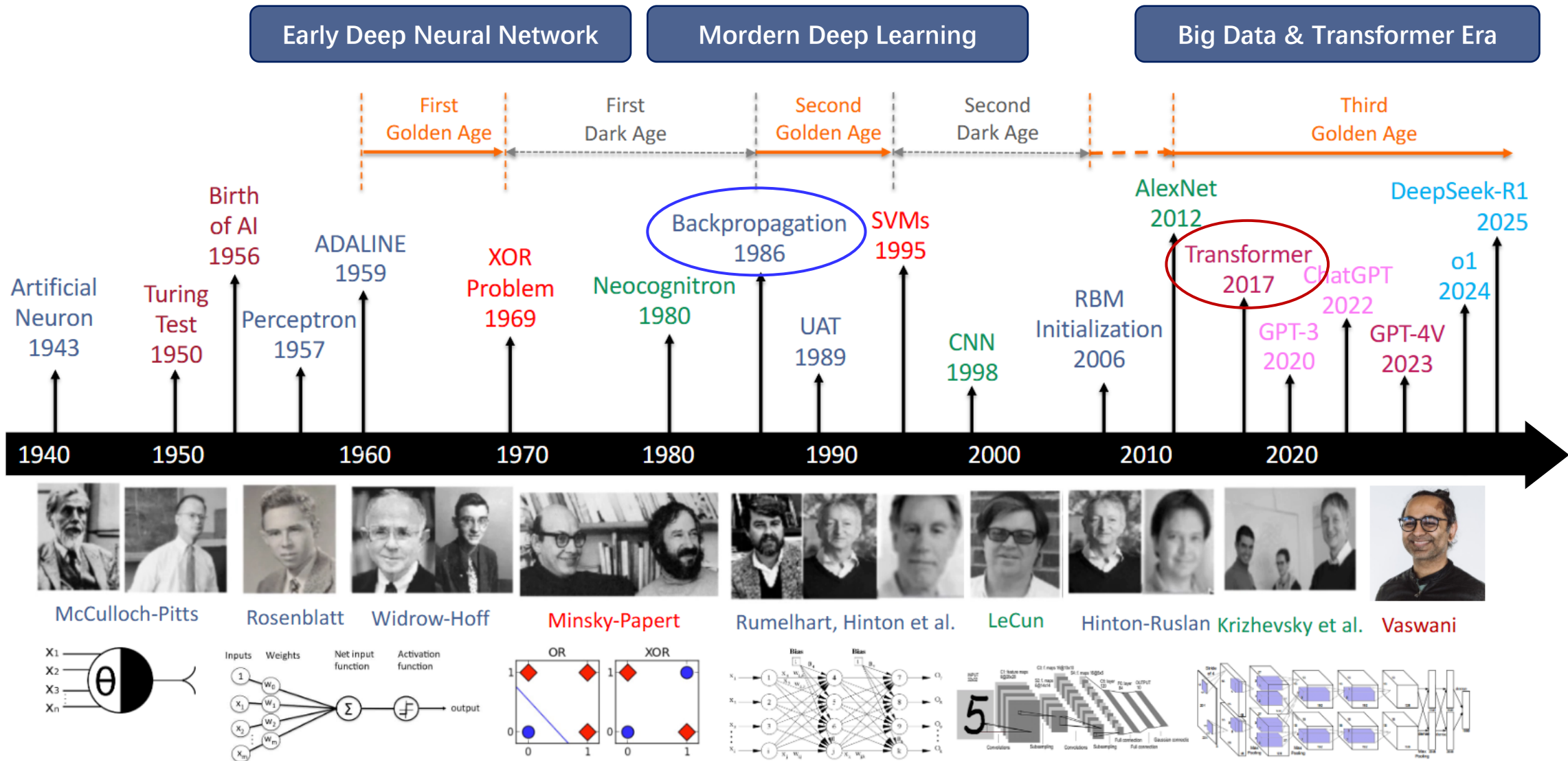
19%

Output

How Neural Network Learns?



Given a number of **input-output examples**, **(supervised) learning** aims to **find the optimal weights** that can map any new data to the correct output

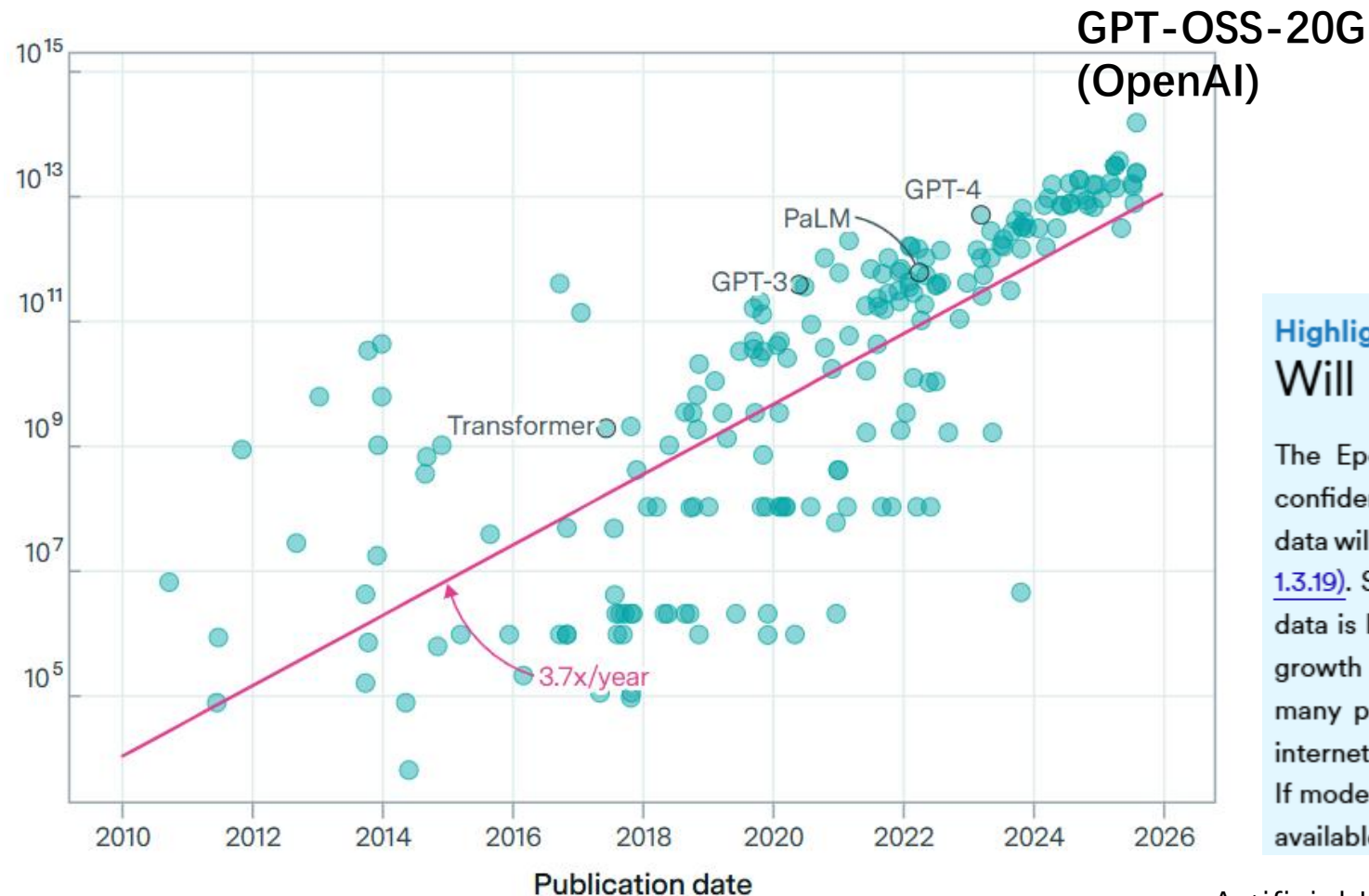


<https://medium.com/@edmond.po/a-brief-history-of-ai-with-deep-learning-26f7948bc87b>

Deep learning has become the dominant foundation for AI development, with increasingly capable models emerging

AI's Hidden Cost

Language model dataset grows at 3.7x per year

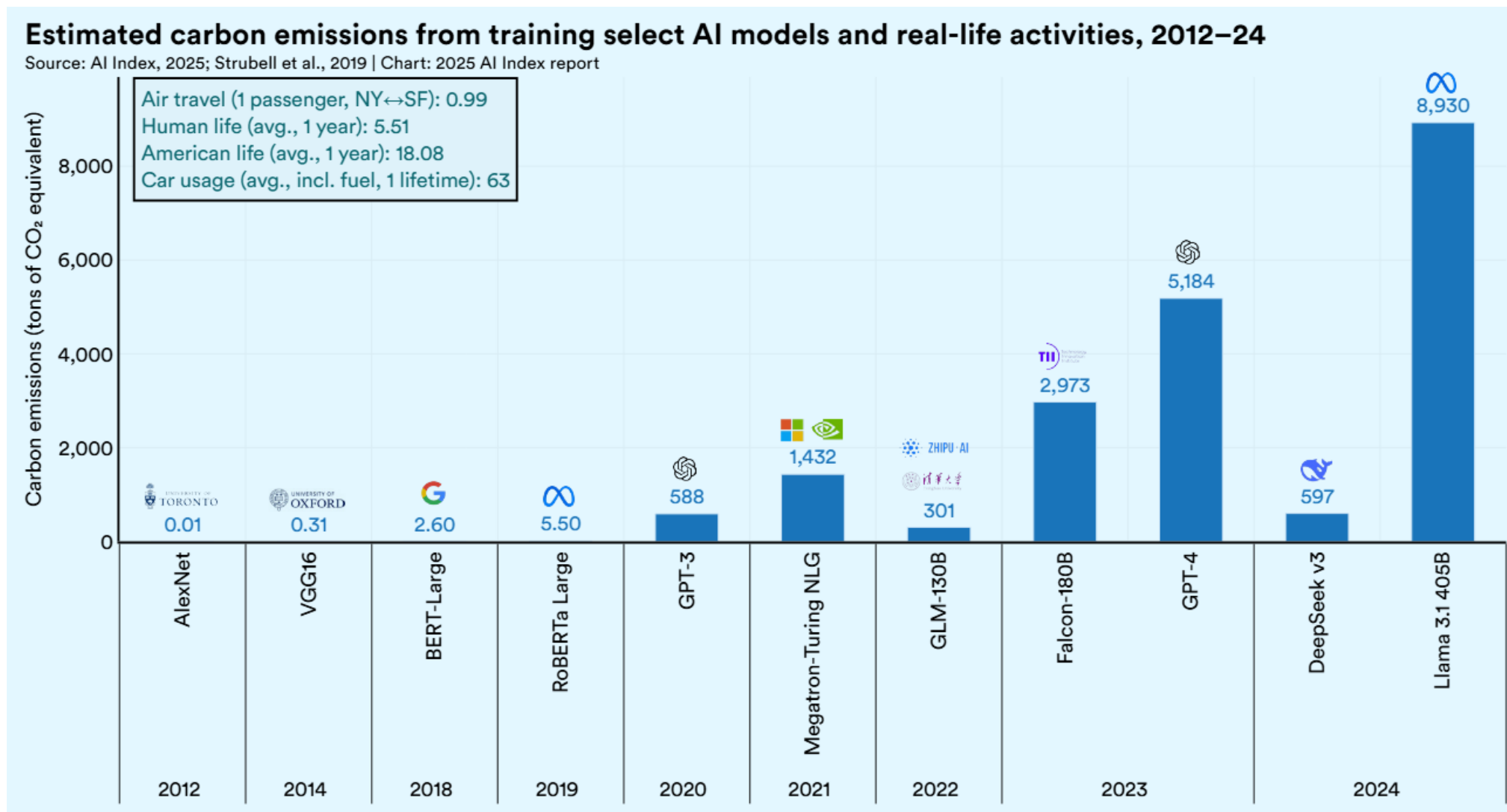


Highlight:

Will Models Run Out of Data? (con

The Epoch AI research team projects, with an 80% confidence interval, that the current stock of training data will be fully utilized between 2026 and 2032 ([Figure 1.3.19](#)). Several factors influence the point in time when data is likely to run out. One key factor is the historical growth of dataset sizes, which depends on how many people generate and contribute content to the internet. Another important factor is computer usage. If models are trained in a compute-optimal manner, the available data stock can last longer. However, if models

AI's Hidden Cost



Most existing AIs are static learners

Learn (train) once, then deploy (inference)

How is AI Learning Different from Human Learning ?



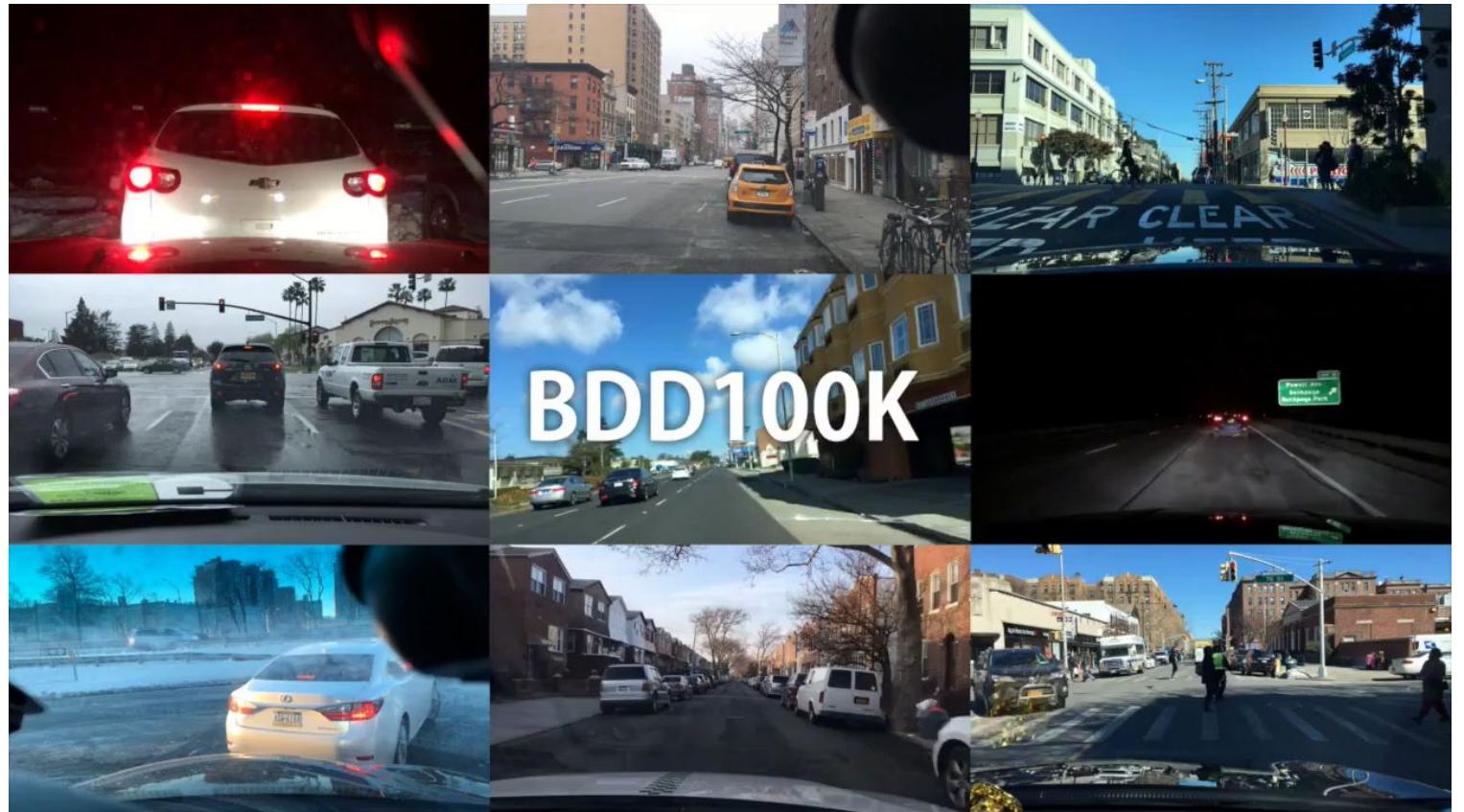
What characteristics of human learning are not easy for AI?

Adaptability

The ability to adapt the model to new situation and environments

Driving perception example

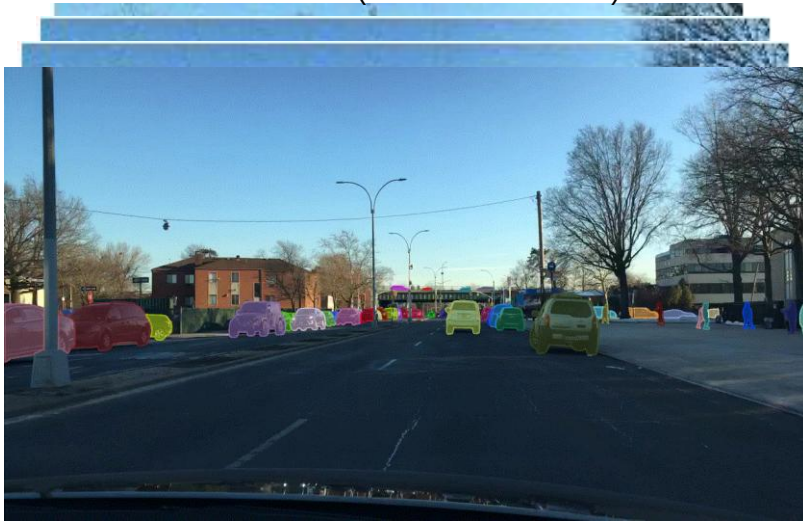
- Weather
- Lighting
- City type
- Road condition
- ...



Adaptability Challenge in AI

BD100K (United States)

Training
Data
(labeled)



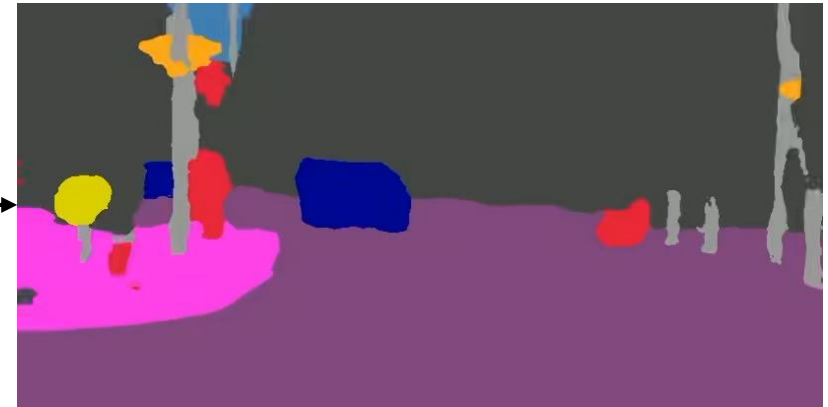
CityScape (Germany)

Testing
Data



Neural
Network
SegFormer
(2022)

Directly apply a model to a
different environment is
prone to error!



Adaptability Challenge in AI



"Under certain circumstances, the [autonomous vehicle] may enter and drive at speed in freeway construction zones due to inappropriately prioritizing the avoidance of other freeway hazards and/or failing to recognize the construction zone," the recall notice states.

Waymo Recalls Nearly 4,000 Robotaxis After Driverless Cars Enter Closed Highway Construction Zones



L.A. food delivery robot attempts to deliver in flooded streets

"My team and I walked down the stairs to check out the rain and noticed the Coco Robot zip by quickly and by the time we got by the door and looked outside, it was struggling, trying to get up the curb," she said.

Adaptability Challenge in AI

- AI models are **only reliable within the confine of the training scenarios**
- How to handle new scenarios?
 - Collect more data
Data collection is costly, sometimes not feasible!
 - Train different models for each scenario
Not scalable! AI training and operation has significant energy, time and labor cost!

Standard supervised learning paradigm lacks efficiency & scalability

Transfer Learning

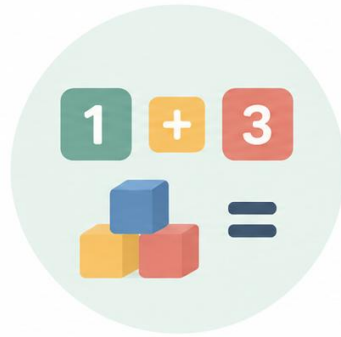
How to adapt an AI model trained on one (or more) source domains to a target domain, often without label or with very few labels

How AI Learning Differs from Human Learning ?

2. Acquire new knowledge & skills continuously through the lifetime



reading



arithmetic



drawing



sports

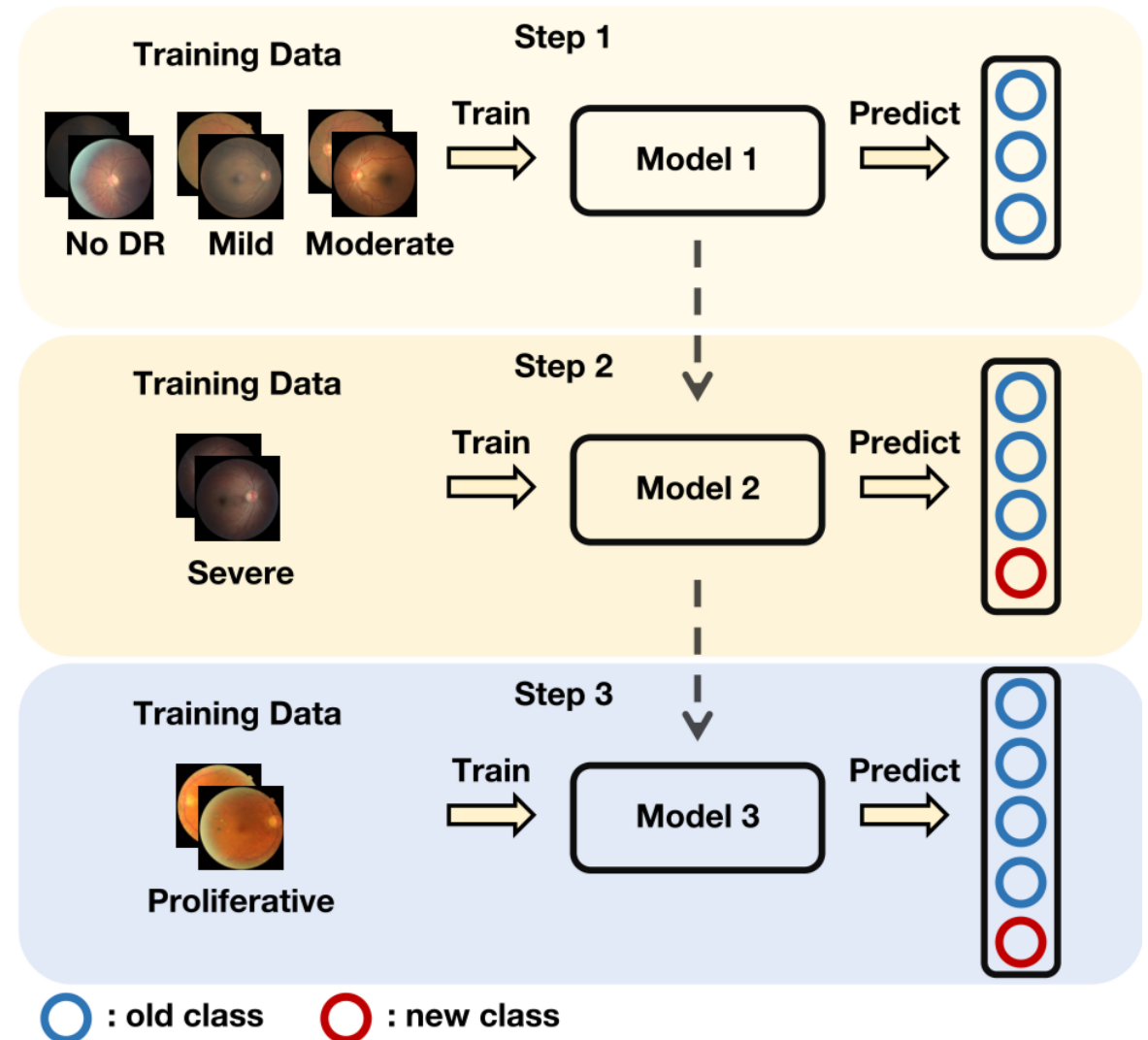
- Learning one thing makes you good at learning other things
- Do not forget old skills when new skill is learned

Incremental Learning Challenge in AI

Learn new knowledge or tasks
progressively from a stream of data

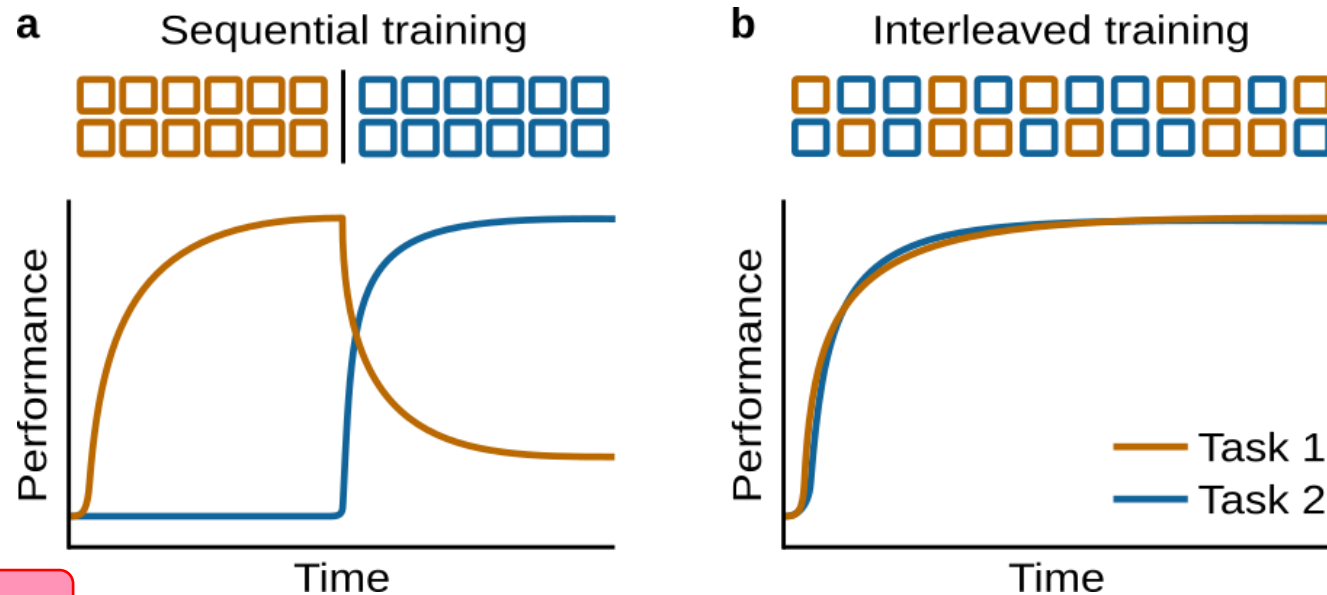
- Naïve solution
 - Mix old and new data, then retrain
- Continue training on the same model using new training data

May risk forgetting previous tasks



Catastrophic Forgetting

- When a neural network is sequentially trained on two tasks, it **rapidly and drastically forgets** the first task while training on the second one.

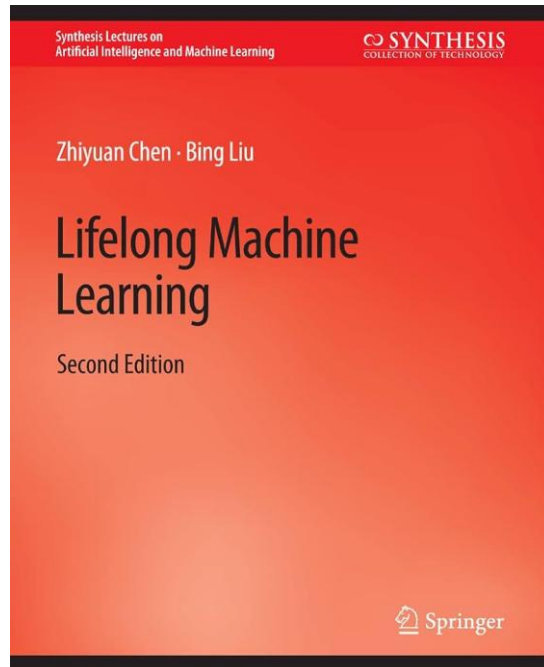


Continual Learning

How to incrementally learn from stream of data of different tasks (environment), such that future tasks can be learned efficiently and without forgetting old tasks.

What is lifelong learning?

- The ability of a model to continuously acquire, adapt, and retain knowledge from new data, tasks, or environments over time.



Course Schedule

- See website syllabus: <https://yangli-feasibility.com/home/classes/aie2026fall/>
- Next class: Training and evaluating a neural network (Lab)
- What to prepare:
 - Form groups of two people
 - Get familiar with <https://teachablemachine.withgoogle.com/> and prepare data/project ideas